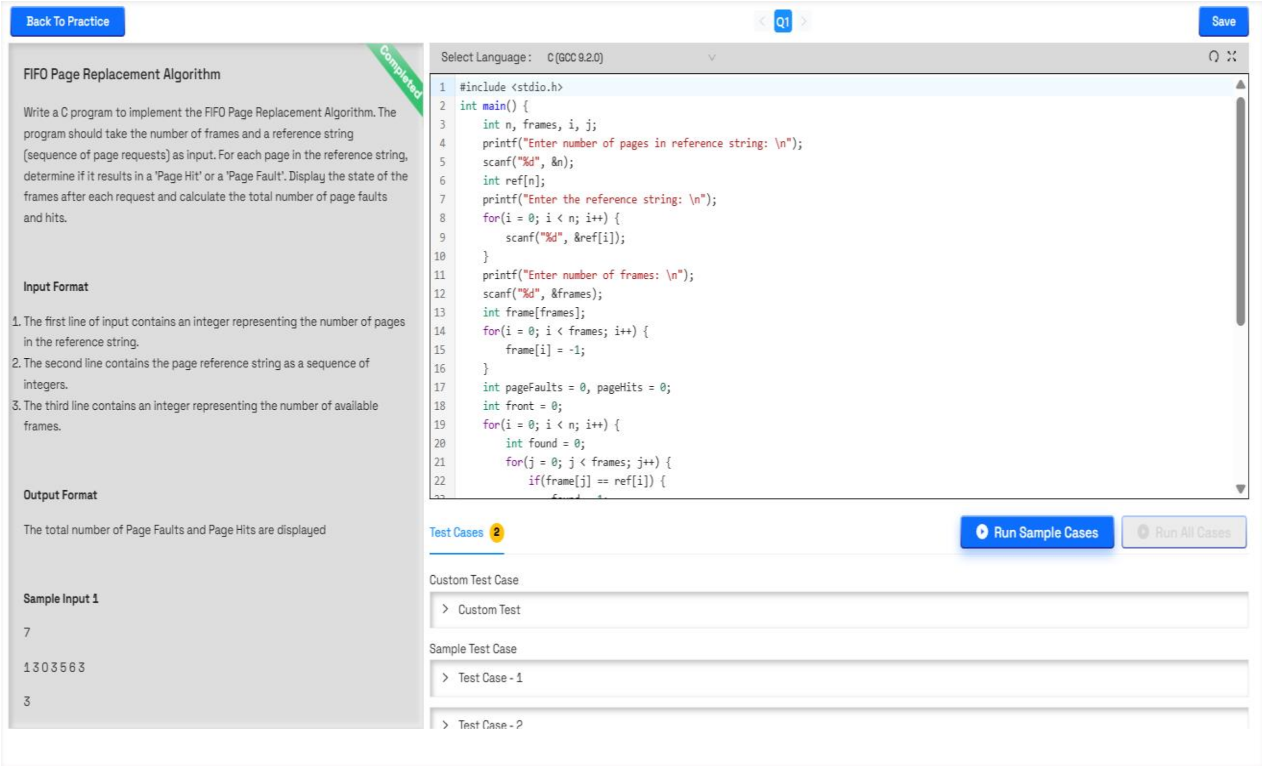
Expt 11a: FIFO Page Replacement:



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FIFO Page Replacement Algorithm

Write a C program to implement the FIFO Page Replacement Algorithm. The program should take the number of frames and a reference string (sequence of page requests) as input. For each page in the reference string, determine if it results in a 'Page Hit' or a 'Page Fault'. Display the state of the frames after each request and calculate the total number of page faults and hits.

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string.
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The total number of Page Faults and Page Hits are displayed

Sample Input 1

7

1303563

3

Select Language: C (GCC 9.2.0)

```c
#include <stdio.h>
int main() {
    int n, frames, i, j;
    printf("Enter number of pages in reference string: \n");
    scanf("%d", &n);
    int ref[n];
    printf("Enter the reference string: \n");
    for(i = 0; i < n; i++) {
        scanf("%d", &ref[i]);
    }
    printf("Enter number of frames: \n");
    scanf("%d", &frames);
    int frame[frames];
    for(i = 0; i < frames; i++) {
        frame[i] = -1;
    }
    int pageFaults = 0, pageHits = 0;
    int front = 0;
    for(i = 0; i < n; i++) {
        int found = 0;
        for(j = 0; j < frames; j++) {
            if(frame[j] == ref[i]) {
```

Test Cases 2 | Run Sample Cases | Run All Cases

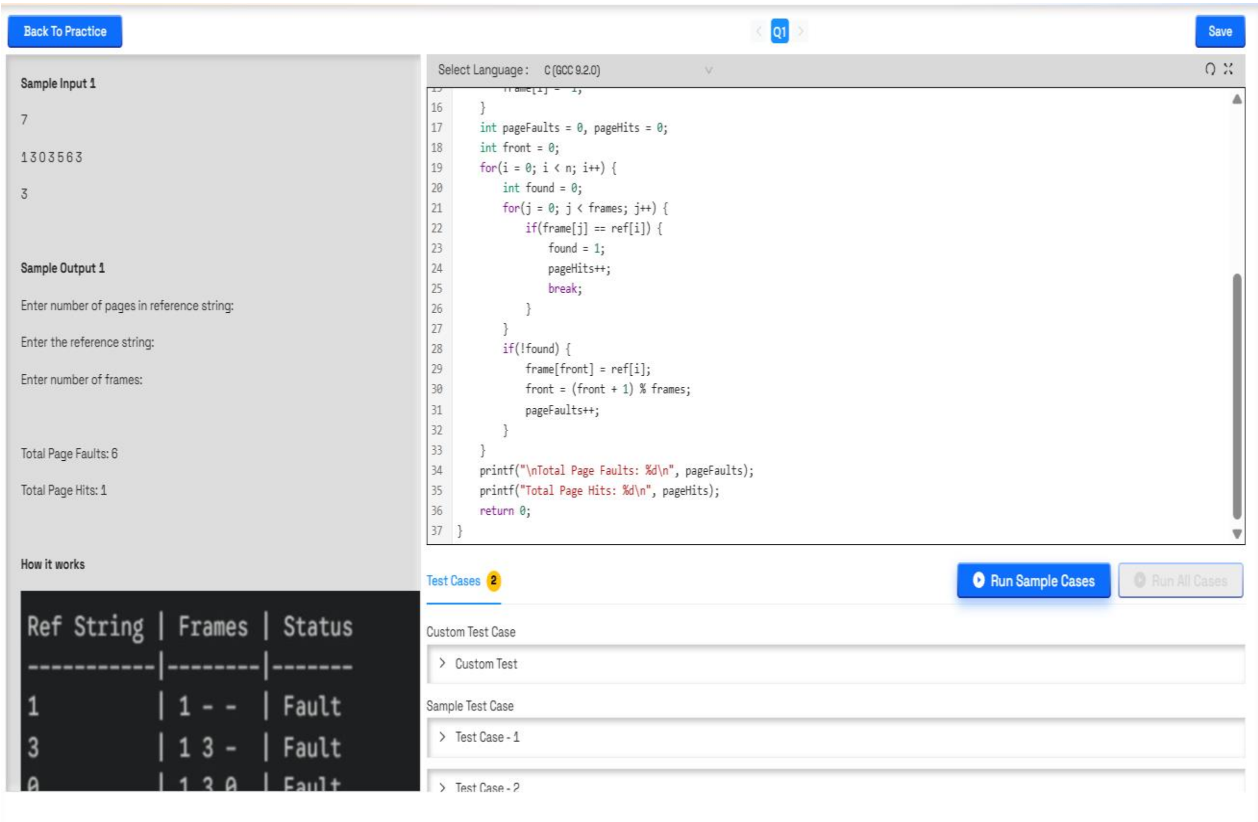
Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

> Test Case - 2



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Sample Input 1

7

1303563

3

Sample Output 1

Enter number of pages in reference string:

Enter the reference string:

Enter number of frames:

Total Page Faults: 6

Total Page Hits: 1

How it works

Ref String	Frames	Status
1	1 - -	Fault
3	1 3 -	Fault
0	1 3 0	Fault

Select Language: C (GCC 9.2.0)

```c
    }
    int pageFaults = 0, pageHits = 0;
    int front = 0;
    for(i = 0; i < n; i++) {
        int found = 0;
        for(j = 0; j < frames; j++) {
            if(frame[j] == ref[i]) {
                found = 1;
                pageHits++;
                break;
            }
        }
        if(!found) {
            frame[front] = ref[i];
            front = (front + 1) % frames;
            pageFaults++;
        }
    }
    printf("\nTotal Page Faults: %d\n", pageFaults);
    printf("Total Page Hits: %d\n", pageHits);
    return 0;
}
```

Test Cases 2 | Run Sample Cases | Run All Cases

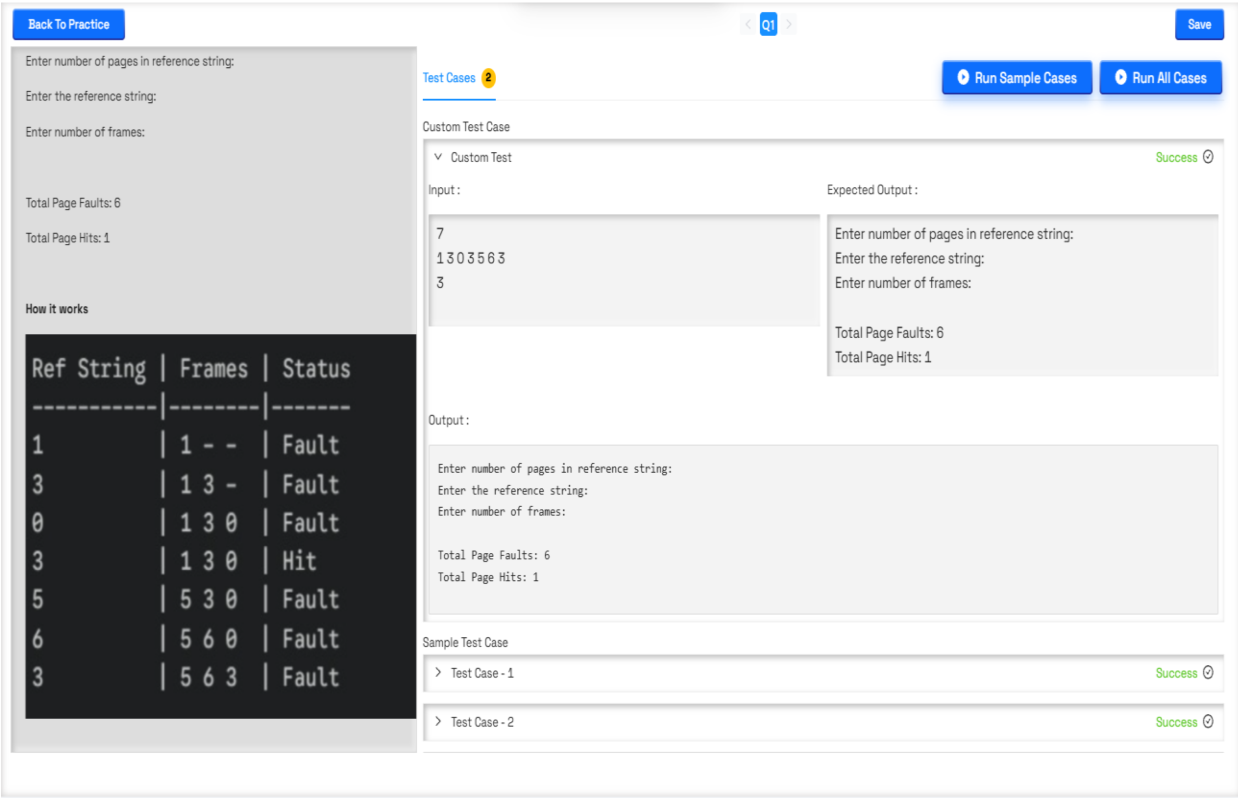
Custom Test Case

> Custom Test

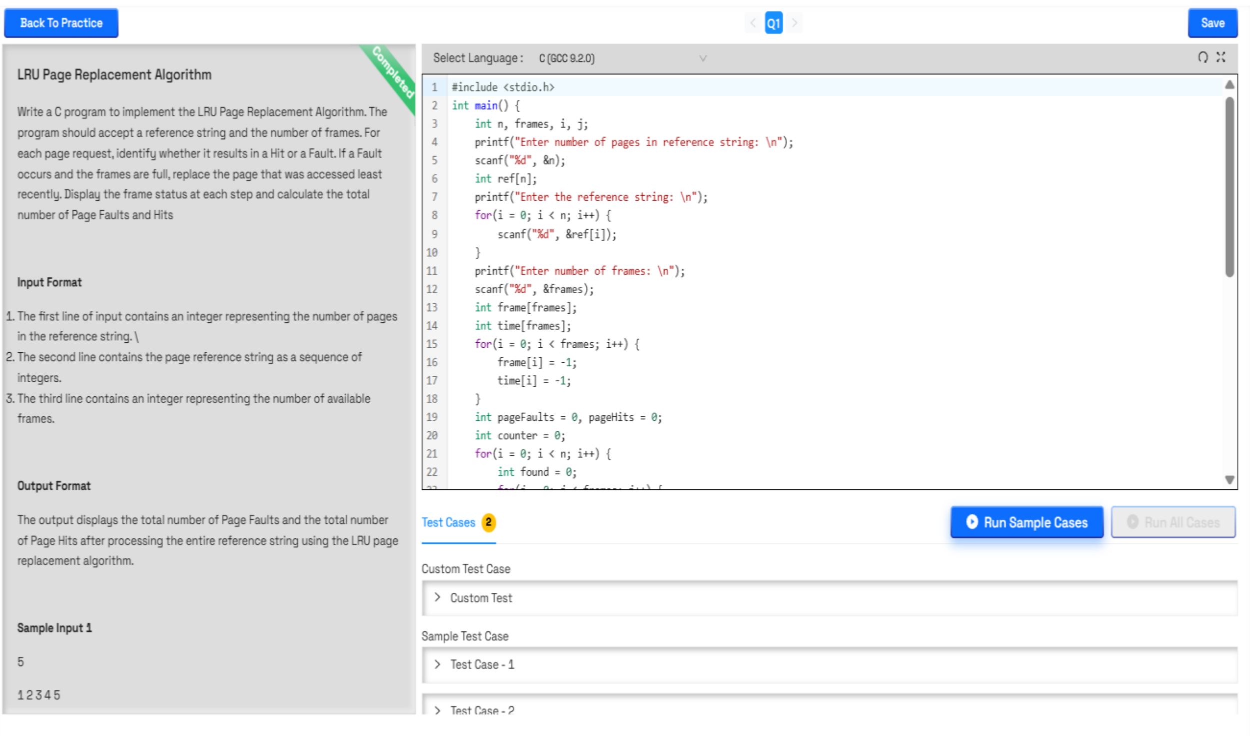
Sample Test Case

> Test Case - 1

> Test Case - 2



Expt 11b: LRU Page Replacement:



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Sample Input 1

5

1 2 3 4 5

5

Sample Output 1

Enter number of pages in reference string:

Enter the reference string:

Enter number of frames:

Total Page Faults: 5

Total Page Hits: 0

How it works

Step	Page	Frame State	Hit / I
1	1	1 - - - -	Fault
2	2	1 2 - - -	Fault
3	3	1 2 3 - -	Fault

Select Language : C (GCC 9.2.0)

```
            if(frame[j] == ref[i]) {
                found = 1;
                pageHits++;
                time[j] = counter++;
                break;
            }
        }
        if(!found) {
            int lru_index = 0;
            for(j = 1; j < frames; j++) {
                if(time[j] < time[lru_index]) {
                    lru_index = j;
                }
            }
            frame[lru_index] = ref[i];
            time[lru_index] = counter++;
            pageFaults++;
        }
    }
    printf("\nTotal Page Faults: %d\n", pageFaults);
    printf("Total Page Hits: %d\n", pageHits);
    return 0;
}
```

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test Case

Custom Test

Sample Test Case

Test Case - 1

Test Case - 2

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Save

Enter number of pages in reference string:

Enter the reference string:

Enter number of frames:

Total Page Faults: 5

Total Page Hits: 0

How it works

Step	Page	Frame State	Hit / I
1	1	1 - - - -	Fault
2	2	1 2 - - -	Fault
3	3	1 2 3 - -	Fault
4	4	1 2 3 4 -	Fault
5	5	1 2 3 4 5	Fault

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test Case

Custom Test — Success

Input :

```
5
1 2 3 4 5
5
```

Expected Output :

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 5
Total Page Hits: 0
```

Output :

```
Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 5
Total Page Hits: 0
```

Sample Test Case

Test Case - 1 — Success

Test Case - 2 — Success